Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories & Story Points)

Date: 26 June 2026

Team ID: XXXXX

Project Name: Smart Lender

Maximum Marks: 5 Marks

1. Product Backlog

Backlog ID	Feature (Epic)	Priority
PB-1	Create Home Page	High
PB-2	Design Loan Prediction Form	High
PB-3	Data Preprocessing	High
PB-4	Train Random Forest Model	High
PB-5	Save Model (rdf.pkl)	High
PB-6	Develop Flask Backend	High
PB-7	Connect Frontend with Backend	High
PB-8	Display Prediction Result	High
PB-9	Test the Application	Medium
PB-10	Documentation and Deployment	Medium

2. Sprint Planning

Sprint 1 – Requirement Analysis & Design (Days 1–2)

Tasks

- Understand project requirements.
- Study the loan prediction dataset.
- Design the project architecture.
- Create UI wireframes.

Deliverables

- Requirement document

- System design
- Project plan

Sprint 2 – Model Development (Days 3–5)

Tasks

- Clean and preprocess the dataset.
- Encode categorical values.
- Train the Random Forest model.
- Evaluate model accuracy.
- Save the trained model as **rdf.pkl**.

Deliverables

- Trained Machine Learning model
- Model evaluation report

Sprint 3 – Web Application Development (Days 6–8)

Tasks

- Develop HTML pages.
- Design CSS.
- Create Flask backend.
- Connect the frontend with the Machine Learning model.
- Display prediction results.

Deliverables

- Functional Smart Lender web application

Sprint 4 – Testing & Deployment (Days 9–10)

Tasks

- Perform system testing.
- Fix bugs.
- Verify prediction results.
- Prepare documentation.
- Deploy the project.

Deliverables

- Final application
- Project documentation
- Deployment-ready solution

3. User Stories & Story Points

Story ID	User Story	Story Points
US-1	As a user, I want to access the Smart Lender homepage.	2
US-2	As a user, I want to enter my loan application details.	5
US-3	As a user, I want the system to validate my inputs.	3
US-4	As a user, I want an instant loan approval prediction.	5
US-5	As a bank officer, I want accurate loan predictions to support lending decisions.	3
US-6	As an administrator, I want the application to run reliably without errors.	2

Total Story Points = 20

4. Team Velocity

- **Sprint Duration:** 10 Days
- **Total Story Points:** 20

Average Velocity

$$\text{Average Velocity} = \frac{\text{Total Story Points Completed}}{\text{Total Number of Sprints}}$$

5. Burndown Chart (Sample Data)

Day	Remaining Story Points
Day 1	20
Day 2	18
Day 3	16
Day 4	13
Day 5	10

Day	Remaining Story Points
Day 6	8
Day 7	6
Day 8	4
Day 9	2
Day 10	0

Project Milestones

Milestone	Expected Outcome
Requirement Analysis	Finalized project scope and requirements
Model Development	Trained and saved Random Forest model (rdf.pkl)
Web Development	Completed Flask application with HTML and CSS
Integration	Connected frontend with Machine Learning model
Testing	Verified prediction accuracy and application functionality
Deployment	Final Smart Lender application ready for demonstration